Lab 9: Random Variables

STAT 141, Week 11

Your name here

In this lab, you will

1. Practice working with the Binomial distribution conceptually and in `R`
2. Calculate properties of continuous random variable distributions

Add any collaborators here!

I collaborated with…

Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

Exercises

Exercise 1: Arachnopobia and the Binomial Distribution

A Gallup Poll found that 7% of teenagers (ages 13 to 17) suffer from arachnophobia and are extremely afraid of spiders. At a summer camp there are 10 teenagers sleeping in each tent. Assume that these 10 teenagers are independent of each other.

a. Let X be the number of teenagers in a tent of 10 teenagers who suffer from arachnophobia. Precisely explain why X follows a Binomial(n,p) distribution, and state what values n and p are. [Only text is needed for this question]

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b. Using what we've learned about the Binomial distribution, calculate E(X), the expected number of teenagers in a tent who suffer from arachnophobia. [Only text is needed for this question]

c. Using R, calculate the probability that *at least one* teenager suffers from arachnophobia. (*Hint: remember that the R functions `pbinom()` and `dbinom()` can help you calculate quantities related to Binomial distributions.*) [Only code is needed for this question]

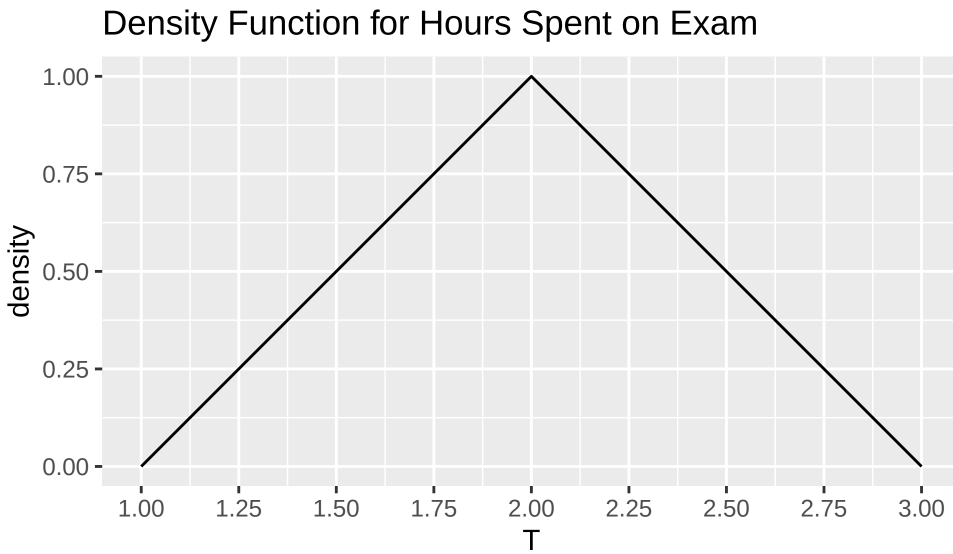
d. Using R, calculate the probability that exactly 2 of them suffer from arachnophobia. (*Hint: remember that the R functions `pbinom()` and `dbinom()` can help you calculate quantities related to Binomial distributions.*) [Only code is needed for this question]

e. Using R, calculate the probability that at most 1 of them suffers from arachnophobia. (*Hint: remember that the R functions `pbinom()` and `dbinom()` can help you calculate quantities related to Binomial distributions.*) [Only code is needed for this question]

f. If the camp counselor wants to make sure no more than 1 teenager in each tent is afraid of spiders, does it seem reasonable for him to randomly assign teenagers to tents? [Only text is needed for this question]

Exercise 2: Practice with Continuous Random Variables

Suppose the number of hours a particular student takes to complete a statistics exam is modeled by a continuous random variable T with the following density function:



Models of this type arise frequently in settings where we have sparse information to model a continuous phenomenon (i.e. we may only have estimates for the minimum, maximum, and most likely values for the phenomenon).

As a reminder, the probability that T falls within some range will be the area under the density function for that range (e.g., $P(1.5 \leq T \leq 2)$ will be the area under the density function from 1.5 to 2). For each of the following parts of the problem, use *geometric reasoning* to compute areas (*you do not need to use calculus to answer any of these problems*).

a. Verify that the function plotted above indeed has area of 1. **Your explanation should be mathematical in nature, but please explain your thought process in words.** [Only text is needed for this question]

b. What is the probability that the student takes at most 2.5 hours to complete the exam? **Explain your reasoning.** [Only text is needed for this question]

c. During what interval of length 1 does the student have the largest probability of completing the exam. That is, find a number a so that $P(a < T < a+1)$ is as large as possible. **Explain your reasoning.** [Only text is needed for this question]

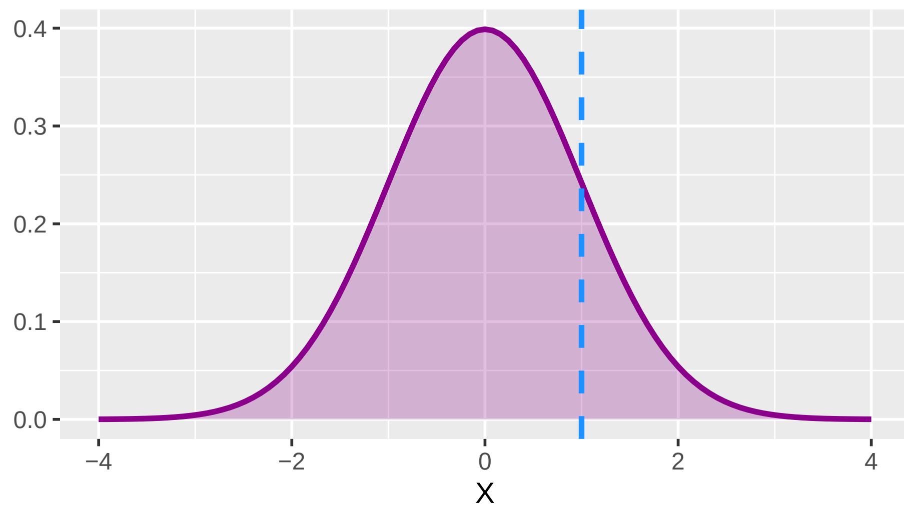
Exercise 3: Probabilities and Quantiles

For Exercise 1, we can use `pbinom()` to easily calculate probabilities like $P(X \leq k)$ when X follows a *Binomial* distribution. We can (and very soon, often will!) use `R` to make similar calculations for continuous random variable distributions, like the *Normal* distribution.

`R` follows similar naming conventions when working with different random variable distributions. If X follows a Normal distribution and we want to calculate a **cumulative probability** $P(X \leq q)$, we can use `pnorm(q, mu, sigma)` (instead of `pbinom`, which was for Binomial random variables), where `mu` and `sigma` are the mean and standard deviation parameters. For now, let's assume that `mu = 0` and `sigma = 1`, which are the default settings in `R` when we just call `pnorm(q)`.

a. The chunk below visualizes a Normal random variable X (with `mu = 0` and `sigma = 1`), and a vertical line representing $X = 1$.

```
ggplot(data.frame(x = c(-4, 4)), aes(x)) +
  stat_function(fun = dnorm, geom = "area", fill = "magenta4", alpha = 0.25) +
  stat_function(fun = dnorm, color = "magenta4", lwd = 1) +
  geom_vline(aes(xintercept = 1), color = "dodgerblue", lwd = 1, lty = 2) +
  labs(title = "", x = "X", y = "")
```



In a new chunk below, use `R` to calculate $P(X \leq 1)$, i.e. the area to the *left* of the blue dashed line. [Only code is needed for this question]

b. Below, play around with calculating probabilities for various points under the Normal distribution above. What happens when you use smaller versus larger values of `q` in `pnorm(q)`? How should we think about the graph above changing when we change `q`? [Code and text is needed for this question]

c. Soon in this class we'll be very interested in calculating **quantiles of theoretical distributions** like this one for inference purposes. For example, we might want to find the 0.95 quantile, or the point q where 95% of the area under the curve is to the left of q. Lucky for us, `R` has functions to help us do this for several types of theoretical distributions. For this normal distribution, `qnorm(p)` will return the **quantile** (on the x-axis) of a given **cumulative probability** p (it will give us q such that $P(X \leq q) = p$ for a given p).

Use `qnorm()` to calculate the **0.85** quantile of our Normal distribution above. Write a statement interpreting the value. Does it make sense to you that this value is close to the location of the blue dashed line above? [Code and text is needed for this question]

d. Using the `ggplot` above as a reference, on the board, sketch a picture visualizing the quantity returned by `pnorm(-2)`. [You don't need to record a response here]

e. Using the `ggplot` above as a reference, on the board, sketch a picture visualizing the quantity returned by `qnorm(0.5)`. [You don't need to record a response here]